

Herlock Rahimi

Ph.D. Candidate, Electrical & Computer Engineering, Yale University

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SUMMARY

Ph.D. candidate in machine learning and optimization, seeking a **research internship**. I design learning algorithms with provable guarantees and implement them in PyTorch: federated and distributed learning under heterogeneous, partially available clients; distributionally robust and constrained learning; and optimal-transport and gradient-flow methods for sampling and generative modeling. Seven papers and preprints, including **ICASSP 2026** and **IEEE CAMSAP 2025** (Student Best Paper Finalist). Strong mathematical background (dual B.Sc. in Computer Engineering and Pure Mathematics; national olympiad medalist).

EDUCATION

Yale University

New Haven, CT

Ph.D. in Electrical & Computer Engineering (Ph.D. Candidate since Nov 2025)

2023 – Present

M.Phil. (May 2026) · M.Sc. (Mar 2025), Electrical & Computer Engineering

Advisors: Dionysis Kalogerias, Amin Karbasi · Graduate Top-Up Fellowship for highest overall coursework GPA (2023, 2024)

Sharif University of Technology

Tehran, Iran

B.Sc. in Computer Engineering (GPA 19.01/20) & B.Sc. in Pure Mathematics (GPA 19.34/20)

2018 – 2023

Top 10% of graduating class

RESEARCH EXPERIENCE

Federated Learning under Client Availability & Drift

Yale University · 2024 – Present

Advisor: Dionysis Kalogerias

- Proved the first convergence guarantee for **Agnostic FedAvg** under random, non-uniform client availability, at rate $O(1/\sqrt{T})$ without knowledge of the participation distribution (IEEE CAMSAP 2025, Best Student Paper Finalist).
- Designed **FedAVOT**, an aggregation rule that casts federated averaging as masked optimal transport between availability and importance distributions, solved by Sinkhorn scaling; proved convergence and showed large gains in low-availability regimes (ICASSP 2026).
- Developed **FedeRage**, a provably convergent agnostic FL method under general client drift; extended the framework to risk-averse FL with CVaR and distributionally robust objectives.
- Built PyTorch and cvxpy simulation pipelines for heterogeneous, non-IID federated environments.

Constrained Learning, Duality & Sampling Dynamics

Yale University · 2024 – Present

- Showed when Lagrangian (primal-dual) training is justified for constrained learning (fairness, safety, risk constraints) beyond convexity, via zero duality gap for universally learnable concept classes.
- Sole-authored a weighted SDE that implements the **Wasserstein-Fisher-Rao gradient flow**, coupling optimal transport and information geometry for faster sampling (Yale AI Symposium lightning talk).

Reasoning in Large Language Models

Yale University

With Amin Karbasi and Zhuoran Yang

- Studying chain-of-thought reliability, search-augmented inference, and curriculum-based alignment; developing evaluation protocols for structured reasoning and compositional generalization.

Generative & Causal Models for Biomedical Data

Collaborative projects · 2025 – 2026

- LEO-CVAE: entropy-guided loss weighting and uncertainty-aware sampling for conditional VAE oversampling; outperformed SMOTE-style and generative baselines on ADNI and TCGA lung-cancer data.
- Causal longitudinal prior-fitted networks (transformer-based) for counterfactual outcome prediction.

Undergraduate Research

Sharif University of Technology · 2020 – 2023

- Attention-based reinforcement learning agents for visual question answering (with H. Rabiee).
- Graph neural networks for histopathology classification with robustness and explainability (with M. H. Rohban).
- B.Sc. thesis: symmetry-invariant representation learning inspired by supersymmetry (with M. H. Rohban).

INDUSTRY EXPERIENCE

Research Intern / Data Scientist, AI Med

Tehran, Iran · Jun – Sep 2022

- Developed and evaluated deep learning models for breast-cancer detection from histopathology images using real hospital data (Imam Khomeini Hospital); built attention-based CNN classifiers with explainability metrics.

PUBLICATIONS & PREPRINTS

1. **H. Rahimi**, D. Kalogerias. FedeRage: Provably Convergent Agnostic Federated Learning under General Client Drift. *arXiv:2609.21057*, 2026.
2. **H. Rahimi**, S. Pougkakiotis, D. Kalogerias. Constrained Learning with Universally Learnable Concept Classes. *arXiv:2608.08414*, 2026.
3. A. Zare, A. Zare, **H. Rahimi**, R. Salarikia, M. Kashkooli. Causal Longitudinal Prior-Fitted Networks for Counterfactual Outcome Prediction. *arXiv:2606.05797*, 2026.
4. **H. Rahimi**, D. Kalogerias. FedAVOT: Exact Distribution Alignment in Federated Learning via Masked Optimal Transport. *IEEE ICASSP*, 2026.
5. **H. Rahimi**. Weighted Stochastic Differential Equation to Implement Wasserstein–Fisher–Rao Gradient Flow. *arXiv:2512.17878*, 2025.
6. A. Zare, A. Zare, P. S. Pezeshki, **H. Rahimi**, A. Ebrahimi, I. Vázquez-García, L. A. Celi. Uncertainty-Aware Generative Oversampling Using an Entropy-Guided Conditional Variational Autoencoder. *arXiv:2509.25334*, 2025.
7. **H. Rahimi**, D. Kalogerias. Convergence of Agnostic Federated Averaging. *IEEE CAMSAP*, 2025. **Student Best Paper Finalist.**

TECHNICAL SKILLS

Programming	Python (PyTorch, TensorFlow, cvxpy), C++, Java, Julia, R, SQL, MongoDB, Bash, Git
Machine Learning	Federated & distributed learning, generative models (VAEs, diffusion), graph neural networks, reinforcement learning, LLM reasoning & evaluation, causal inference
Mathematics	Convex & nonconvex optimization, stochastic approximation, duality, optimal transport, information geometry, statistical learning theory, high-dimensional probability

HONORS & TALKS

- Student Best Paper Finalist, IEEE CAMSAP 2025.
- Talks: Yale AI Symposium lightning talk (2026); AI Symposium, Hartford (2026); IEEE CAMSAP (2025).
- Graduate Top-Up Fellowship, Yale University (2023, 2024).
- Top-3 Finalist, Iranian National Mathematics Olympiad for College Students (2022) and Statistics Olympiad (2021).
- Ranked 9th nationwide, Iranian University Entrance Exam (2018); Silver Medal, Iranian National Mathematical Olympiad (2017).

TEACHING & SERVICE

- **Instructor:** Information Geometry (Yale, 2025); Chaos, Fractals, and Noise (Yale, 2026); online Machine Learning course (Sharif, 100+ students).
- **Teaching Assistant, Yale:** Advanced Stochastic Processes, Statistical Learning Theory, Applied ML & Causal Inference, Deep Learning on Graph-Structured Data, Intro to Computer Engineering.
- **Teaching Assistant, Sharif:** 10+ courses incl. ML Theory, High-Dimensional Probability, Artificial Intelligence (Head TA), Algorithms, Compilers.
- **Reviewer:** NeurIPS 2025 Workshop on Reliable ML from Unreliable Data.
- **Languages:** English (fluent), Persian (native).